Functional Analysis

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August 2023

Contents

This is notes for the one semester course on Functional Analysis taking place in IISER KOLKATA.

Topics that will be covered in the course are as follows:

i. Banach spaces and linear maps on Banach spaces.

ii. Hahn Banach theorems, namely, Hahn Banach Extension and Hahn Banach Separation theorems.

iii. Baire Category theorem: Uniform boundedness principle, open mapping theorem and closed graph theorem.

iv. Hilbert spaces (generalization of inner product spaces).

v. Compact operator and spectrum of compact and self-adjoint operator.

vi. Weak and Weak* topology (in this topic one of the main results will be the generalization of Bolzano-Weierstrass theorem).

1 Introduction

In linear algebra, we usually deal with finite dimensional vector spaces and linear operators between them. However, in functional analysis we study continuous linear maps $F : X \to Y$, where X and Y are infinite dimensional normed linear spaces. When we say X is a normed linear space, we mean that we are not only considering the vector space structure of X but also the topological structure that is induced by the norm defined on X. For example, $\mathbb{R}^N$'s are normed linear spaces where they are equipped with the usual vector space structure and the norm is the Euclidean norm.

On further advancement in this field, people also extend their study to non-linear maps between the infinite dimensional normed linear spaces, but, in this course, we will restrict our discussion to the linear case only. Linear functional analysis has applications in linear PDEs, approximation theory, numerical analysis, etc., where mostly researchers try to figure out solutions to equations of the form $F(x) = y$, where $x \in X$, $y \in Y$, F is an operator and X and Y are as defined before. One of the ways that these can be solved is that first we find their solutions in the finite dimensional case and under some constraints, we pass to the limit to obtain solutions in the infinite dimensional case.

2 Some fundamental spaces

Definition 2.0.0.1. ***(Topological vector space).*** A topological vector space is a vector space V that is endowed with a Hausdorff topology such that the maps

$$(x, y) \in V \times V \mapsto x + y \in V \text{ and } (\alpha, x) \in \mathbb{F} \times V \mapsto \alpha x \in V$$

are continuous, where each product space is equipped with the product topology coming from the topology on V and the usual topology on the scalar field $\mathbb{F}$.

Definition 2.0.0.2. ***(Normed linear space).*** A normed linear space is a vector space V endowed with a norm; that is, it is a topological vector space where the topology is induced by the norm defined on it.

Notation: We will denote the normed linear spaces as $(V, \|.\|)$.

If $(V, \|.\|)$ is a normed linear space, then V becomes a metric space with the metric d defined by $d(x, y) = \|x-y\|$, for all $x, y \in V$ and the topology induced by this metric is called the norm topology.

Note that if $\{x_n\}$ is a sequence in V then $x_n \to x$ in V iff $\|x_n \to x\| \to 0$.

Definition 2.0.0.3. ***(Banach space).*** A normed linear space $(V, \|.\|)$ is called a Banach space if V is a complete metric space with respect to the metric d defined as above.

Following is an example of a normed linear space which is not a Banach pace.

Example 2.0.0.4. Let V be the space of all real polynomials in $[0,1]$, where the norm is defined as

$$\|p\| = \max_{x\in[0,1]} |p(x)|,$$

where p is any polynomial in V. It follows from the Stone Weierstrass theorem that V is not a Banach space, because any arbitrary continuous function which is not a polynomial does not belong to V.

Notation: For $\mathbb{R}^N = \{x = (x_i)_{1\leq i\leq N} : x_i \in \mathbb{R}\}$, $\|.\|_p$ will denote the l_p norm, that is for $x \in V$, $\|x\|_p = (\sum_{i=1}^{n} |x_i|^p)^{1/p}$.

Remark 2.0.0.5. $\|.\|_p$ where $0 < p < 1$ is not a norm. One can check it for $p = 1/2$ in $\mathbb{R}^2$ by considering $x = (1,9)$, $y = (4,16)$ and showing that the triangle inequality is not satisfied.

Definition 2.0.0.6. ***(Sequence space).*** Consider the set of all real or complex sequences $x = (x_1, x_2, \dots)$. Let $1 \leq p < \infty$. We define the space

$$l_p = \{x : \sum_{i=1}^{\infty} |x_i|^p < \infty\}.$$

This is a vector space where the vector addition and scalar multiplication is defined pointwise.

For $x \in V$, define

$$\|x\|_p = (\sum_{i=1}^{\infty} |x_i|^p)^{1/p}.$$

The above defines a norm on l_p.

Similarly, we can define

$$l_\infty = \{x = (x_i) : \sup_{1\leq i<\infty} |x_i| < \infty\},$$

which is the space of all bounded sequences. This too clearly defines a vector spaces under componentwise addition and scalar multiplication. On this space we define the following norm: For $x \in V$,

$$\|x\|_\infty = \sup_{1\leq i<\infty} |x_i|.$$

Lemma 2.0.0.7. *$l_p \subset l_\infty$.*

Proof. Let $x = (x_1, x_2, \dots) \in l_p$. Note that for $p \geq 1$, $(\sup_{1\leq i<\infty}|x_i|)^p = \sup_{1\leq i<\infty}|x_i|^p$ and since, $\sup_{1\leq i<\infty}|x_i|^p \leq \sum_{i=1}^{\infty} |x_i|^p$, raising both sides to $1/p$, it follows that $x \in l_\infty$. ☐

2.1 Important Banach spaces

Theorem 2.1.0.1. *Let $1 \leq p < \infty$. Then, l_p is a Banach space.*

Proof. We just need to show that l_p is a complete space. To show that, let us consider a Cauchy sequence $x^{(n)} = \{(x_i^n)_{i\in I} : n \in \mathbb{N}\}$ in l_p. Let $\epsilon > 0$. Then, there exists $N(\epsilon) \in \mathbb{N}$ such that for each i,

$$\textstyle\sum_{i=1}^{\infty} (|x_i^{(n)} - x_i^{(m)}|^p)^{1/p} < \epsilon \;\forall n, m \geq N(\epsilon).$$

Let $k \in \mathbb{N}$. Then,

$$\textstyle\sum_{i=1}^{k} (|x_i^{(n)} - x_i^{(m)}|^p) < \epsilon^p \;\forall n, m \geq N(\epsilon).$$

Since for each i, $\{x_i^{(m)}\}$ is a Cauchy sequence in $\mathbb{R}$, and since $\mathbb{R}$ is complete, $x_i^{(m)} \to x_i$ as $m \to \infty$. Taking $m \to \infty$ in the above inequality, we have,

$$\textstyle\sum_{i=1}^{k} (|x_i^{(n)} - x_i|^p) < \epsilon^p \;\forall n \geq N(\epsilon).$$

Let, $x = (x_i)_{i\in I}$, then since k is arbitrary, it follows that

$$\|x^{(n)} - x\|_p < \epsilon.$$

. Hence, $x^{(n)} \to x$ in l_p. Since, $x^{(n)}$ was arbitrary, every Cauchy sequence in l_p converges in l_p. ☐

Theorem 2.1.0.2. *l_∞ is a Banach space.*

Proof. This can be proved using the same line of thought as in the proof of theorem 2.1.0.1, the only difference being that we need to use l_∞ norm instead of l_p norm. ☐

Note: Given any $x = (x_1, x_2, \dots) \in V$, we can think of x as a function $f : \mathbb{N} \to \mathbb{R}$ where $f(i) = x_i$. Then,

$$\|f\|_\infty = \sup_{i \in \mathbb{N}} |f(i)|.$$

We made this comment because we want to motivate the following example:

Example 2.1.0.3. Let $C[0,1]$ denote the set of all real valued continuous functions on the closed interval $[0,1]$. This is a vector space under pointwise addition and scalar multiplication and we define

$$\|f\| = \sup_{x \in [0,1]} |f(x)| = \max_{x \in [0,1]} |f(x)|.$$

We could replace the suprema by maxima because the function f is defined on a compact interval $[0,1]$, and thus f attains its maxima. It is a routine check that it indeed defines a norm on $C[0,1]$. Moreover, with this norm, $C[0,1]$ is a Banach space, the proof of which is easy, as given any Cauchy sequence $\{f_n\}$ in $C[0,1]$, it's just a matter of manipulation of $\epsilon - \delta$ definitions of uniform convergence and uniform continuity of the sequence of functions $\{f_n\}$ (here, uniform convergence and uniform continuity holds because f_n's are continuous functions on the compact interval $[0,1]$) to proof its convergence to a function $f \in C[0,1]$.

3 Important inequalities

3.1 Hölder's inequality

Theorem 3.1.0.1. *Let $1 < p < \infty$ and p^* be the conjugate exponent of p. Let $x, y \in \mathbb{R}^N$, then,*

$$\sum_{i=1}^{N} |x_i y_i| \leq \|x\|_p \|y\|_{p^*}.$$

Note: For $p = 1$ and $p = \infty$, it follows from the definition, so we do not consider these cases.

Proof. To prove this, we will use generalised A.M.-G.M. inequality which states that: for $a, b \geq 0$ and p, p^* as above,

$$a^{1/p} b^{1/p^*} \leq \frac{a}{p} + \frac{b}{p^*}. \tag{1}$$

Consider $x, y \neq 0$ (as $x = 0$ or $y = 0$ gives the zero equality) and choose $a = \left(\frac{|x_i|}{\|x\|_p}\right)^p$ and $b = \left(\frac{|x_i|}{\|x\|_{p^*}}\right)^{p^*}$. Using the above inequality, the result follows.

Now we prove the generalised A.M.-G.M. inequality, which is obvious if $a, b = 0$, so, WLOG we consider $a \geq b > 0$. Note that $\forall\ t \geq 1$ and $0 < k < 1$, $t^k < k(t-1) + 1$. This can be proved by using basic real analysis techniques. Now, if we choose $t = \frac{a}{b}$, $k = \frac{1}{p}$ (where $1 < p < \infty$) and substitute in the aforementioned inequality, we get

$$\left(\frac{a}{b}\right)^{1/p} \leq \frac{1}{p}\left(\frac{a}{b} - 1\right) + 1.$$

Multiplying both sides of the above inequality by b yields the generalised A.M.-G.M. inequality. ☐

Corollary 3.1.0.2. *Let $x = (x_1, x_2, \dots) \in l_p$ and $y = (y_1, y_2, \dots) \in l_{p^*}$, where p and p^* are conjugate exponents. Then,*

$$\sum_{i=1}^{\infty} |x_i y_i| \leq \|x\|_p \|y\|_{p^*}.$$

Proof. Let $k \in \mathbb{N}$. Then, truncating x and y upto k terms, we get

$$\sum_{i=1}^{k} |x_i y_i| \leq \|(x_1, x_2, \dots, x_k)\|_p \|(y_1, y_2, \dots, y_k)\|_{p^*} \leq \|x\|_p \|y\|_{p^*}.$$

Since, the right hand side is independent of k and the choice of k was arbitrary, taking $k \to \infty$, we obtain our result. ☐

3.2 Minkowaski inequality

Now, we will prove another important inequality, known as Minkowaski inequality, which will be heavily used in this course apart from the Holder's inequality.

Theorem 3.2.0.1. *Let $x, y \in \mathbb{R}^n$ and $1 \leq p \leq \infty$. Then,*

$$\|x+y\|_p \leq \|x\|_p + \|y\|_p.$$

Proof. For the cases $p = 1$ and $p = \infty$, it is very easy to see just by expanding the definitions. So, we move on to the case $1 < p < \infty$:

$$\textstyle\sum_{i=1}^n |x_i + y_i|^p \leq \sum_{i=1}^n |x_i + y_i|^{p-1}|x_i + y_i| \leq \sum_{i=1}^n |x_i + y_i|^{p-1}|x_i| + \sum_{i=1}^n |x_i + y_i|^{p-1}|y_i|.$$

Using Holder's inequality and considering p^* to be the conjugate exponent of p, it follows from the above that

$$\textstyle\sum_{i=1}^n |x_i + y_i|^p \leq (\|x\|_p + \|y\|_p)(\sum_{i=1}^n |x_i + y_i|^{(p-1)p^*})^{1/p^*}.$$

Now, since, p and p^* are conjugate exponents, we have that $p = (p-1)p^*$, then the above inequality implies that $\|x+y\|_p^p \leq (\|x\|_p + \|y\|_p)\|x+y\|_p^{p/p^*}$. If $\|x+y\|_p \neq 0$, then, $\|x+y\|_p^{p-p/p^*} \leq \|x\|_p + \|y\|_p$. But, $p - p/p^* = 1$, hence the Minkowaski inequality follows in this case. Also, in the other case when $\|x+y\| = 0$, the Minkowaski inequality follows trivially, so we are done. □

Corollary 3.2.0.2. *Let $x, y \in l_p$ and $1 \leq p \leq \infty$. Then,*

$$\|x+y\|_p \leq \|x\|_p + \|y\|_p.$$

Proof. We can use the same idea of truncation to prove this corollary, as we did in 3.1.0.2. □

Since, in sequence spaces, $l_p \subset l_\infty$, it is a natural question to ask whether the l_∞ norm is the limit of all l_p norms. It turns out that it is indeed true.

Theorem 3.2.0.3. *Let $x \in \mathbb{R}^n$ be arbitrary. Then,*

$$\lim_{p\to\infty} \|x\|_p = \|x\|_\infty.$$

Note: The proof for the sequence spaces follows by truncation and using this theorem.

Proof. Let $\|x\|_\infty = \max\limits_{1\leq i\leq n} |x_i| = |x_{i_0}|$ (say). Now, by using Minkowaski's inequality,

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} = \left(|x_{i_0}| + \sum_{i\neq i_0} |x_i|^p\right)^{1/p} = |x_{i_0}| \left(1 + \frac{\sum_{i\neq i_0}|x_i|^p}{|x_{i_0}|^p}\right)^{1/p}.$$

As for each $i = 1, 2, \ldots, n$, $\frac{|x_i|}{|x_{i_0}|} \leq 1$, it follows that as $p \to \infty$ $\|x\|_p \to \|x\|_\infty$, and hence the result. □

4 Linear algebra on infinite dimensional normed linear spaces

Till date, the linear algebra that we have learnt, dealt with finite dimensional vector spaces (usually over $\mathbb{R}$ or $\mathbb{C}$) and linear transformations between them. One of the advantages of finite dimensional normed vector spaces is that if we consider any linear transformation between them (here, the finite dimensionality of the domain is enough, the codomain can be of any dimension), then the continuity of that linear transformation comes for free. But it is not so in the case of infinite dimensional normed vector spaces, as it can be proved that given any infinite dimensional Banach space, there exists a discontinuous linear functional on it. Henceforth, when we are working with linear transformations with domain being a infinite dimensional normed linear space, which we will often do in this course, it is essential to consider those linear transformations which are also continuous, otherwise they might fail to satisfy some nice properties which usually linear transformations between finite dimensional normed vector spaces do.

To see why any linear transformation with domain being a finite dimensional normed vector space is continuous, let us consider a linear map $T : (\mathbb{R}^n, \|.\|_1) \to (\mathbb{R}^m, \|.\|_1)$. It is enough to show for this T because the proof that we are going to provide remains true even if the codomain is an infinite dimensional normed linear space over $\mathbb{R}$.

Let $\{e_i\}$ be the standard basis of $\mathbb{R}^n$ and let $x \in \mathbb{R}^n$ be arbitrary. Then, $x = \sum_{i=1}^n x_i e_i$, which implies $T(x) = \sum_{i=1}^n x_i T(e_i)$, where for each $i = 1, 2, \ldots, n$, $T(e_i) \in \mathbb{R}^m$. Now, if we let $M = \max\limits_{1\leq i\leq n} \|T(e_i)\|$, it follows that $\|T(x)\| \leq M\|x\|_1$. Then, for any $y \in \mathbb{R}^n$, by linearity of T, considering the element $x - y \in \mathbb{R}^n$, we have that $\|T(x) - T(y)\| \leq M\|x - y\|_1$. Thus, T satisfies the Lipschitz condition and hence, it is continuous.

Now, we prove an important theorem.

Theorem 4.0.0.1. *Let $T : (V, \|.\|_V) \to (W, \|.\|_W)$ be a linear map. Then, the following are equivalent:*

i. $T : (V, \|.\|_V) \to (W, \|.\|_W)$ *is continuous.*

ii. T *is continuous at* 0 *in* V*.*

iii. $\exists$ *a constant* $c > 0$ *such that* $\|T(x)\|_W \leq c\|x\|_V$*, for all* $x \in V$*.*

iv. Let $B = \{x \in V : \|x_V\| \leq 1\}$*, then,* $T(B)$ *is bounded in* W*.*

Proof. $(i) \implies (ii)$ is obvious.

$(ii) \implies (i)$ Considering a sequence $x_n \to x$ in V, we have that $x_n - x \to 0$ in V. Applying T both sides, (i) follows by the linearity of T and the sequential criteria of continuity.

$(iii) \implies (ii)$ If $x_n \to 0$ in V, then, by (iii) it follows that $T(x_n) \to 0$ in W. Thus, by sequential criteria, T is continuous at 0.

$(ii) \implies (iii)$ Since, T is continuous at $x = 0$, for $\epsilon = 1$, $\exists\, \delta > 0$ such that whenever $\|x\|_V < \delta$, $\|T(x)\|_W < 1$. Considering $y = \frac{x}{\|x\|}\frac{\delta}{2}$, we have $\|y\|_V < \delta$ which implies $\|T(y)\|_W < 1$ and thus, $\|T(x)\|_W < \frac{2}{\delta}\|x\|_V$. If we let $c = 2/\delta$, the result follows.

$(iii) \implies (iv)$ is obvious as from (iii) it follows that $T(B) \subset B(0, c)$.

$(iv) \implies (iii)$ For $x = 0$, the inequality is trivial. So, let $x(\neq 0) \in V$. Define $y = \frac{x}{\|x\|}$. Then, $\|y\|_V = 1$ implies $y \in B$ and thus $\exists\, r > 0$ such that $\|T(y)\|_W < r$. It follows that $\|T(x)\|_W < r\|x\|_V$, for all $x \in V$. ☐

Remark 4.0.0.2. Due to condition (iv) in the above theorem, continuous linear transformations are also called bounded linear maps. So, to prove or disprove the continuity of a certain linear transformation, (iv) can be used.

Definition 4.0.0.3. Let $T : (V, \|.\|) \to (W, \|.\|)$ be a continuous linear transformation. The operator norm on T is defined as

$$\|T\|_{L(V,W)} = \sup_{\substack{x\in V\\ \|x\|\leq 1}} \|T(x)\|_W$$

Here,

$$L(V, W) = \{T \mid T : V \to W \text{ is a continuous linear transformation}\}.$$

It is not difficult to show that $L(V, W)$ is a vector space and indeed $\|.\|_{L(V,W)}$ is a norm on it.

4.1 Examples of continuous and discontinuous linear transformations

Example 4.1.0.1. Let $T : (\mathbb{R}^n, \|.\|_1) \to (\mathbb{R}^m, \|.\|_2)$ be any linear transformation. If $\{e_i\}_{i=1}^n$ is basis of $\mathbb{R}^n$, then for any $x \in \mathbb{R}^n$, we have that $\|T(x)\|_2 \leq k\|x\|_1$, where $k = \max\limits_{1\leq i\leq n} \{\|T(e_i)\|_2\}$. From 4.0.0.1 (iii), it follows that T is continuous.

Example 4.1.0.2. Let us consider the linear transformation

$$T : l_2 \to l_2$$
$$(x_1, x_2, \dots, x_n, \dots) \mapsto \Big(\frac{x_1}{1}, \frac{x_2}{2}, \dots, \frac{x_n}{n}, \dots\Big)$$

If $x = (x_1, x_2, \dots, x_n, \dots) \in l_2$, then,

$$\begin{aligned}\|T(x)\|_2 &= \left(x_1^2 + \frac{x_2^2}{4} + \dots + \frac{x_n^2}{n^2} + \dots\right)^{1/2} \\ &\leq (x_1^2 + x_2^2 + \dots x_n^2 + \dots)^{1/2} = \|x\|_2.\end{aligned}$$

So, considering $c = 1$, from 4.0.0.1 (iii), it follows that T is continuous.

Example 4.1.0.3. Let p and p^* be conjugate exponents. Consider $y = (y_1, \dots, y_n, \dots) \in l_{p^*}$. Define

$$T_y : l_p \to (\mathbb{R}, \|.\|_1)$$
$$x \mapsto \sum_{i=1}^{\infty} x_i y_i$$

Then, $\|T_y(x)\|_1 = |\sum_{i=1}^{\infty} x_i y_i| \leq \|x\|_p \|y\|_{p^*}$ by Hölder's inequality. So, considering $c = \|y\|_{p^*}$, from 4.0.0.1 (iii), it follows that T is continuous.

Example 4.1.0.4. Let us consider the linear transformation

$$T : C[0,1] \to (\mathbb{R}, \|.\|)$$
$$f \to \int_0^1 f(x)dx$$

where $C[0,1]$ is equipped with the sup norm, that is for $f \in C[0,1]$, $\|f(x)\| = \max\limits_{x\in[0,1]} |f(x)|$.

Now, $\|T(f)\|_1 = |\int_0^1 f(x)dx| \leq \|f\|_{C[0,1]}$. Again, considering $c = 1$, we have that T is continuous.

Remark 4.1.0.5. From the above example, it is an immediate observation that any linear transformation $T : L^1(0,1) \to (\mathbb{R}, \|.\|_1)$ is continuous.

Example 4.1.0.6. Consider the linear transformation

$$T : (C[0,1], \|.\|_\infty) \to (C[0,1], \|.\|_\infty)$$
$$f \mapsto \frac{df}{dx}$$

If T is continuous, then, $\exists\ c > 0$ such that

$$\|T(f)\|_\infty \leq c\|f\|_\infty$$

that is,

$$\max_{x\in[0,1]} \left|\frac{df(x)}{dx}\right| \leq c \max_{x\in[0,1]} |f(x)|. \tag{2}$$

Considering the sequence of function $f_n(x) = x^n$, where $x \in [0,1]$, from the above inequality we would get that $n \leq c$ for all $n \geq 1$, which is a contradiction. Thus, T is not continuous.

Remark 4.1.0.7. Observe that the above linear transformation can be made continuous by considering the following two norms (it is not difficult to see once we compare with the inequality (2)), which we name $\|.\|_\alpha$ and $\|.\|_\beta$ respectively:

i. $\|f\|_\alpha = \|f\|_\alpha + \|f'\|_\alpha$.

ii. $\|f\|_\beta = \max\limits_{x\in[0,1]}\{\|f\|_\infty, \|f'\|_\infty\}$.

Note that for both the cases $c = 1$ will work.

Example 4.1.0.8. Let us consider the following linear transformation from $C[0,1]$ to $\mathbb{R}$ which takes $f \in C[0,1]$ to the evaluation of the derivative map of f at the point 0, that is,

$$T : (C[0,1], \|.\|_\infty) \to \mathbb{R}$$
$$f \to f'(0)$$

Considering the sequence of functions $f_n(x) = (1-x)^n$, we again get that T is not continuous wrt the l_∞ norm, but is indeed continuous with the two norms $\|.\|_\alpha$ and $\|.\|_\beta$ mentioned in the above remark.

Example 4.1.0.9. Now, we consider the linear transformation that sends f to its evaluation at the point 0, but here here we take $C[0,1]$ equipped with the l_1 norm, that is, for $f \in C[0,1]$, $\|f\|_1 = \int_0^1 |f(x)|dx$. So, we define

$$T : (C[0,1], \|.\|_1) \to (\mathbb{R}, \|.\|)$$
$$f \mapsto f(0)$$

If T is continuous, the, $\exists\, c > 0$ such that

$$\|T(f)\| \leq c\|f\|_1$$

that is,

$$|f(0)| \leq c \int_0^1 |f(x)|dx. \tag{3}$$

But, it is quite obvious that it cannot hold for every continuous function in $[0,1]$. Example of one such function is:

$$f_n : [0,1] \to \mathbb{R}$$
$$x \mapsto \begin{cases} 1 - nx & \text{if } 0 \leq x \leq \frac{1}{n} \\ 0 & \text{if } \frac{1}{n} \leq x \leq 1. \end{cases}$$

Since, $\int_0^1 f_n(x)dx = \frac{1}{2n}$, from (3) we have that $2n \leq c$, for all $n \geq 1$, which is a contradiction.

Question: Is $(C[0,1], \|.\|_1)$ a Banach space?

Ans. We have seen earlier that $C[0,1]$ with the l_∞ norm is a Banach space. But, it is not so when we equip $C[0,1]$ with the l_1 norm. Considering the sequence of functions in the above example, we have that

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{otherwise.} \end{cases}$$

which is clearly not a continuous function as f is discontinuous at the point $x = 0$. Hence, $f \notin C[0,1]$, which implies $(C[0,1], \|.\|_1)$ is a normed linear space which is not a Banach space.

But, this is not a good example because we are in the Lebesgue setting, and observing that the the limit function $f(x)$ has 0 as its only point of discontinuity, we can redefine the function $f(x)$ to make it continuous. So, to disprove that $(C[0,1], \|.\|)$ is a Banach space, we need to consider a Cauchy sequence in $C[0,1]$ which is L^1 convergent to a function having a positive measure set as its set of discontinuity.

Consider the sequence of functions for $n \in \mathbb{N}$:

$$f_n : [0,1] \to \mathbb{R}$$
$$x \mapsto \begin{cases} 1 & \text{if } x \in [0, 1/2] \\ 1 - (n+1)(x - \frac{1}{2}) & \text{if } x \in [1/2, 1/2 + 1/(n+1)] \\ 0 & \text{if } x \in [1/2 + 1/(n+1), 1]. \end{cases}$$

Under the L^1-norm, this sequence of functions converge to the following limit function:

$$\begin{aligned} f : [0,1] &\to \mathbb{R} \\ x &\mapsto \begin{cases} 1 & \text{if } x \in [0,1/2] \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

and clearly, f does not have removable singularity, so, $f \notin C[0,1]$. □

In similar lines we can show that $C[-1,1]$ equipped with L^1-norm is not a Banach space by considering the following sequence of functions:

$$\begin{aligned} f_n : \mathbb{R} &\to \mathbb{R} \\ x &\mapsto \begin{cases} 0 & \text{if } x \in [-1,0] \\ nx & \text{if } x \in (0,1/n] \\ 1 & \text{if } x \in (1/n,1]. \end{cases} \end{aligned}$$

We show that $\{f_n\}$ is a Cauchy sequence in $\|.\|_1$.

Let $n > m$. Then,

$$\|f_n - f_m\|_1 = \int_{-1}^{1} |f_n - f_m| dx = \int_0^{1/n} |f_n - f_m| dx + \int_{1/n}^{1/m} |f_n - f_m| dx$$

Note that

$$\int_0^{1/n} |f_n - f_m| dx = \int_0^{1/n} (nx - mx) dx = \frac{n-m}{2n^2} \leq \frac{1}{2n},$$

which tends to 0 as $n \to \infty$. Again,

$$\int_{1/n}^{1/m} |f_n - f_m| dx = \int_{1/n}^{1/m} (1 - mx) dx = \frac{n-m}{2n^2} \leq \frac{1}{2n} = \frac{1}{n} \times \left(1 - \frac{m}{2}\right)\left(\frac{1}{m} - \frac{1}{n}\right),$$

which also tends to 0 as $n \to \infty$.

It is easy to observe that under L^1-norm, this sequence of functions $\{f_n\}$ converges to the limit function

$$\begin{aligned} f : \mathbb{R} &\to \mathbb{R} \\ x &\mapsto \begin{cases} 0 & \text{if } x \in [-1,0] \\ 1 & \text{if } x \in (0,1] \end{cases} \end{aligned}$$

which has a jump discontinuity at the point 0, hence $f \notin C[-1,1]$.

Remark 4.1.0.10. The examples have a same line of flavour as that of how it is showed that $(0,1]$ is not a complete metric space wrt the $|.|$ norm by considering the Cauchy sequence $\{\frac{1}{n}\}$ converging to 0.

Note that all the limit functions in the above examples, though fail to be continuous, are L^1 over $[0,1]$, and indeed as we have seen in our previous measure theory course,

i. $\overline{C[0,1]}^{\|.\|_1} = L^1(0,1)$

ii. $\overline{(0,1]}^{|.|} = [0,1]$.

4.2 The operator norm of continuous linear transformations

As one might have seen in their analysis courses, if $T : (V, \|.\|_V) \to \|.\|_W)$ is a continuous linear transformation, then the sup norm of T is defined to be

$$\|T\|_{L(V,W)} = \sup_{\|x\|_V \leq 1} \|T(x)\|_W.$$

Now, the next theorem will provide alternate definitions to the definition of sup norm.

Theorem 4.2.0.1. *Let $T : (V, \|.\|_V) \to \|.\|_W)$ be a continuous linear transformation, then,*

$$\|T\|_{L(V,W)} = \underset{\|x\|_V=1}{sup} \|T(x)\|_W = \underset{x\neq 0}{sup} \frac{\|T(x)\|_W}{\|x\|_V} = inf\{k \in \mathbb{R} : \|T(x)\|_W \leq k\|x\|_V\}.$$

Proof. For simplicity, let us define

i. $\alpha = \sup\limits_{\|x\|_V=1} \|T(x)\|_W$,

ii. $\beta = \sup\limits_{x\neq 0} \frac{\|T(x)\|_W}{\|x\|_V}$,

iii. $\gamma = \inf\limits_{k>0} S$, where, $S = \{k : \|T(x)\|_W \leq k\|x\|_V\}$.

Our goal is to show: $\|T\| = \alpha = \beta = \gamma$.

($\alpha = \beta$): We have $\beta = \sup\limits_{x\neq 0} \frac{\|T(x)\|_W}{\|x\|_V} = \sup\limits_{x\neq 0} \left\| T\left(\frac{y}{\|y\|_V}\right)\right\|_W = \sup\limits_{\|x\|_V=1} \|T(x)\|_W$, where $y = \frac{x}{\|x\|_V}$.

($\beta \leq \gamma$) : For any $\tilde{k} \in S$, we have, $\|T(x)\|_W \leq \tilde{k}\|x\|_V$, that is, $\sup\limits_{x\neq 0} \frac{\|T(x)\|_W}{\|x\|_V} \leq \tilde{k}$. Hence, by definition, $\beta \leq \tilde{k}$, which implies $\beta \leq \inf S$, and therefore, $\beta \leq \gamma$.

($\gamma \leq \beta$) : From definition of β, we have for all $x \in V$, $\|T(x)\|_W \leq \beta\|x\|_V$. So, $\beta \in S$ and since, $\gamma \in S$, it follows that $\gamma \leq \beta$.

Therefore, $\alpha = \beta = \gamma$. Now, if we can just show that $\|T\| = \alpha$, we will be done.

By definition of operator norm, we already have that $\|T\| \geq \alpha$. Now, showing $\alpha \geq \|T\|_{L(V,W)}$ is similar to showing $\gamma \geq \|T\|_{L(V,W)}$. For any $k \in S$, $\|T(x)\|_W \leq k\|x\|_V$. Taking $\sup\limits_{\|x\|_V\leq 1}$ both sides, we have that $\|T\|_{L(V,W)} \leq k\|x\|_V \leq k$. Again, taking $\inf\limits_{k>0}$ both sides, we obtain our required result, that is, $\|T\|_{L(V,W)} \leq \gamma$. □

From the above theorem, since, $\sup\limits_{x\neq 0} \frac{\|T(x)\|_W}{\|x\|_V} = \|T\|_{L(V,W)}$, we can deduce the following corollary:

Corollary 4.2.0.2. *If $T : (V, \|.\|_V) \to \|.\|_W)$ is a continuous linear transformation, then, for all $x \in V$,*

$$\|T(x)\|_W \leq \|T\|_{L(V,W)}\|x\|_V.$$

Let us now explicitly deduce some nice operator norms using the above theorem and corollary.

Example 4.2.0.3. Consider the linear operator

$$T : C[0,1] \to \mathbb{R}$$
$$f \mapsto \int_0^1 f(x)dx.$$

Then, $|T(f)| \leq \|f\|_\infty$. Considering the function $f(x) = 1$, for all $x \in [0,1]$, we have that $\|T(x)\|_\mathbb{R} \leq 1$. Since, $\|T\|_{L(C[0,1],\mathbb{R})} = \gamma = \inf\{k \in \mathbb{R} : \|T(x)\|_\mathbb{R} \leq k\|x\|_{C[0,1]}\}$, it implies $\|T\|_{L(C[0,1],\mathbb{R}} \leq 1$. Again, considering the same function and using $\|T\|_{L(C[0,1],\mathbb{R})} = \alpha$, we obtain $\|T\|_{L(C[0,1],\mathbb{R})} \geq 1$. Thus, $\|T\|_{L(C[0,1],\mathbb{R})} = 1$.

Since, it is getting quite cumbersome, from now on, we will use the notations $\|T\|$ for the operator norm, unless otherwise specified.

Example 4.2.0.4. Consider the linear operator

$$T : C[0,1] \to \mathbb{R}$$
$$f \mapsto f(0).$$

Using the same idea as that in the above example, we can deduce that $\|T\| = 1$.

4.3 Linear integral operator

The linear transformations that we have considered in the above examples are an example of what we want to introduce in this section, namely, linear integral operators. Such operator are often considered over $C(D)$ or $L^p(D)$, where D is a compact subset of $\mathbb{R}^n$.

The general form of a linear integral operator, say, T defined on a closed subset D of $\mathbb{R}$ is

$$T(f(y)) = \int_D K(x,y)f(x)dx,$$

for all $y \in D$. The function $K(x,y)$ is called the *kernel* of this linear integral operator T.

Example 4.3.0.1. *(Fredholm integral operator).* Consider a function $K \in C([0,1]\times[0,1])$ and define the linear operator

$$\begin{aligned} T : C[0,1] &\to C[0,1] \\ f &\mapsto \int_0^1 K(x,y)f(x)dx, \end{aligned}$$

where, K is the kernel of the linear operator T. Note that, by definition of the linear operator T, for every $f \in C[0,1]$, $T(f)$ is a function of y. To prove that T is continuous, we need to show that there exists a constant $C > 0$ such that

$$\|T(f)\|_\infty \leq C\|f\|_\infty.$$

Let us suppose

$$\max_{\substack{x\in[0,1]\\ y\in[0,1]}} |K(x,y)| \leq M.$$

Now,

$$\max_{y\in[0,1]} |T(f)(y)| = \max_{y\in[0,1]} \left| \int_0^1 K(x,y)f(x)dx \right| \leq M\|f\|_\infty,$$

and, therefore, $\|T(f)\|_\infty \leq M\|f\|_\infty$. So, T is a continuous linear transformation. Note that if our kernel is 1, then, we reduce down to our first example, where, $\|T\| = M = 1$. Now, let us see another type of integral operator, called the Volterra integral operator.

If a continuous integral operator is defined on a compact space, then, such an integral operator is called *Fredholm integral operator*, and the corresponding kernel is called *Fredholm kernel.* Clearly, the above linear operator is an example of such an operator.

Example 4.3.0.2. *(Volterra integral operator).* Consider the linear operator

$$\begin{aligned} T : C[0,1] &\to C[0,1] \\ f &\mapsto \int_0^y K(x,y)f(x)dx, \end{aligned}$$

where, $K : C([0,1]\times[0,1]) \to \mathbb{R}$ is the kernel of the integral operator T.

We show that T is a continuous linear operator.

Since, K is continuous on a compact set, it is bounded and uniformly continuous. Let $\epsilon > 0$ be arbitrary. Then, there exists $\delta > 0$ such that for all $y_1, y_2 \in [0,1]$ such that whenever $|y_1, -y_2| < \delta$, $|K(x,y_1) - K(x,y_2)| < \epsilon$, for all $x \in [0,1]$. WLOG, let us assume $y_1 > y_2$ and $\delta \leq \epsilon$. Then, we have

$$\begin{aligned} T(f)(y_1) - T(f)(y_2) &= \int_0^{y_1} K(x,y_1)f(x)dx - \int_0^{y_2} K(x,y_2)f(x)dx \\ &= \int_0^{y_1} (K(x,y_1) - K(x,y_2))f(x)dx + \int_{y_2}^{y_1} K(x,y_2)f(x)dx, \end{aligned}$$

which implies
$$|T(f)(y_1) - T(f)(y_2)| \leq \epsilon \|f\|_\infty y_1 + M\|f\|_\infty \delta = (1+M)\|f\|_\infty \epsilon.$$
Moreover, $\|T(f)(y)\| \leq My\|f\|_\infty \leq M\|f\|_\infty$. Therefore, T is continuous and $\|T\| \leq M$.

Question. Is the space V defined as
$$V = \{f : [0,1] \to \mathbb{R} : \int_0^1 \frac{1}{x} f^2(x)dx < \infty\}$$
closed wrt L^2 norm?

Answer: Let us consider the sequence of functions
$$f_n : \mathbb{R} \to \mathbb{R}$$
$$x \mapsto \begin{cases} nx & \text{if } x \in [0, 1/n] \\ 1 & \text{if } x \in (1/n, 1]. \end{cases}$$
which converges to the limit function
$$f : \mathbb{R} \to \mathbb{R}$$
$$x \mapsto \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x \in (0, 1]. \end{cases}$$
Then, it is easy to check that $\int_0^1 \frac{1}{x} f_n^2(x)dx = \frac{1}{2} + \ln\left(\frac{1}{n}\right)$, which is a finite quantity for each $n \geq 1$. But, $\int_0^1 \frac{1}{x} f^2(x)dx = \int_0^1 \frac{1}{x} \nless \infty$. Hence, V is not closed wrt L^2 norm. □

Note that $V \subset L^2(0,1)$. Moreover, $\overline{V}^{L^2(0,1)} = L^2(0,1)$. It is an easy observation that the limit function f in the above example is a L^2 function in $(0,1)$.

Let us now prove a very important theorem. We define
$$L(V, W) := \{T : V \to W | T \text{ is a continuous linear transformation}\}.$$

Theorem 4.3.0.3. *Let V and W be normed linear spaces. Let W be a Banach space, then, $L(V,W)$ is a Banach space with respect to the operator norm $\|.\|_{L(V,W)}$.*

Proof. Let us consider a Cauchy sequence $\{T_n\}$ in $L(V,W)$. Then, for $\epsilon > 0$, there exists $N(\epsilon) \in \mathbb{N}$ such that
$$\|T_n - T_m\|_{L(V,W)} < \epsilon \ \ \forall n, m \geq N(\epsilon).$$
Since, for each $n \in \mathbb{N}$, T_n is a continuous linear transformation, we have that
$$\|T_n(x) - T_m(x)\|_W < \|T_n - T_m\|_{L(V,W)} \|x\|_V, \tag{4}$$
for all $x \in V$. So, it follows that for each $x \in V$, $T_n(x)$ is a Cauchy sequence in W. Since, W is a Banach space, for all $x \in V$, $T_n(x)$ converges to say, $T(x) \in W$.

We claim that $T_n \to T$ in $L(V,W)$. To prove our claim, we need to show

i. $T \in L(V,W)$, and,

ii. $\|T_n - T\|_{L(V,W)} \to 0$ as $n \to \infty$.

To prove (i), let us first show that T is a linear map. Since, each T_n is a linear map, for all $x, y \in V$, $T_n(x+y) = T_n(x) + T_n(y)$. Taking limit both side, the linearity of T follows. Now, we show that T is continuous. Since, $\{T_n\}$ is a Cauchy sequence, it is bounded, that is, there exists some $M > 0$ such that $\|T_n\|_{L(V,W)} \leq M$, for all $n \in \mathbb{N}$. So, from the continuity of T_n, we have $\|T_n(x)\| \leq M\|x\|_V$, for all $x \in V$, on which after taking limit $n \to \infty$, the continuity of T follows.

To prove (ii), taking $m \to \infty$ in equation (4), we get $\|T_n(x) - T(x)\| \leq \epsilon \|x\|_V$, for all $n \geq N(\epsilon)$, and for all $x \in V$. Since, $\|T_n - T\| = \sup_{\|x\|_V = 1} \|T_n(x) - T(x)\|$, it follows that for all $n \geq N(\epsilon)$, $\|T_n - T\|_{L(V,W)} < \epsilon$, and therefore, $T_n \to T$ in $L(V,W)$. □

Remark 4.3.0.4. The converse of this theorem also holds, which we will prove later using Hahn-Banach theorem.

Corollary 4.3.0.5. *The space of all linear functionals $L(V, \mathbb{R})$ is a Banach space.*

Before dwelling into other important topics of functional analysis, let us see a fun thing! The next little section arose out of the curiosity in knowing the difference between closure and completion of a metric space.

5 Completion of a metric space

Let (X, d_X) be a metric space. Then, (Y, d_Y) is said to be a completion of (X, d_X) if

i. (Y, d_Y) is a complete metric space, and,

ii. there exists an isometry $f : X \to Y$ such that $f(X)$ is dense in Y.

Here, every point of Y is interpreted as an equivalence class of Cauchy sequences of X, where, any two Cauchy sequences, say, x_n and y_n of X are equivalent in Y if $d_Y(x_n, y_n) \to 0$ as $n \to \infty$. Clearly, X is embedded in Y by constant sequences, that is, every element $x \in X$ corresponds to the constant sequence $\{x, x, \dots\}$ in the equivalence class $[x]$ of Y.

In that sense, $\overline{Q} = \mathbb{R}$ and $\overline{(0, 1]} = [0, 1]$, where the isometry f in both the cases is $f(x) = x$.

6 Characterization of finite dimensional spaces

In this section, we will focus on proving two important things:

i. All norms are equivalent in finite dimensional space.

ii. Let $B := \{x \in V : \|x\| = 1\}$ denote the closed unit ball in a normed linear space V. Then B is compact in V iff V is finite dimensional.

The converse of (1) also holds, that is, if V is a normed linear space such that all norms are equivalent, then, V is finite dimensional. We will focus on proving this converse statement after we are done proving (1).

For (2), the following is an example where we consider an infinite dimensional normed linear space and show that the closed ball is not compact.

Example 6.0.0.1. Let us consider the l_2 space. Let e_i denote the element of l_2 which has 1 at its $i - th$ position and 0 at all other places. Then, $\|e_i\|_{l_2} 1$. If the unit ball $\{f : \|f\|_{l_2} \leq 1\}$ is compact, or sequentially compact in l_2, then, the sequence $\{e_i\}$ in l_2 should have a convergent subsequence $\{e_n\}$ in l_2. But, for any $i, j \in \mathbb{N}$, $\|e_i - e_j\|_{l_2} = 2$, hence, the sequence $\{e_i\}$ cannot be Cauchy, that is, in particular, the sequence $\{e_n\}$ cannot converge. Therefore, it follows that the unit ball is not compact in l_2 space.

Before we start with the goals of this section, let us see some definitions and develop some theory which will help in proving them.

Definition 6.0.0.2. Let (X, d) and (X, d') be two metric spaces. We say that d is stronger than d', if for any $x \in X$ and $\epsilon > 0$, there exists $\delta > 0$ such that $B_d(x, \delta) \subseteq B_{d'}(x, \epsilon)$.

Remark 6.0.0.3. By the above definition, it is clear that if a set is open in (X, d'), then, it is open in (X, d). So, at the level of topology, we have that $\tau_{d'} \subseteq \tau_d$.

Note that the two metrics d and d' are said to be *equivalent* if the open sets in both (X, d) and (X, d') are same, that is, topologically, $\tau_d = \tau_{d'}$.

6.1 Equivalence of norms

Lemma 6.1.0.1. *Let $(V, \|.\|)$ and $(V, \|.\|')$ be two normed linear spaces. Then, $\|.\|$ is stronger than $\|.\|'$ iff $\|x\|' \leq \alpha\|x\|$ for all $x \in V$ and $\alpha > 0$.*

Proof. Let d and d' be the metric induced by the norms $\|.\|$ and $\|.\|'$ respectively. Then,

$$d(x, y) = \|x - y\| \quad \text{and} \quad d'(x, y) = \|x - y\|'.$$

Since, d is stronger than d', for $x = 0$ and $\epsilon = 1$, there exits $\delta > 0$ such that $B_d(0, \delta) \subset B_{d'}(0, 1)$. Therefore,

$$\{x \in V : \|x\| < \delta\} \subset \{x \in V : \|x\|' < 1\}.$$

For $x \neq 0$, let us define $y = \frac{\delta}{2}\frac{x}{\|x\|}$. Then, $\|y\| < \delta$, which implies $y \in B_d(0, \delta)$, that is, $\|y\|' < 1$. So, we have $\|x\|' < \frac{2}{\delta}\|x\|$. Considering $\frac{2}{\delta}$ as α, the required result follows. □

Remark 6.1.0.2. The two norms $\|.\|$ and $\|.\|$ are said to be equivalent iff for some $\alpha, \beta > 0$, $\alpha\|x\| \leq \|x\|' \leq \beta\|x\|$.

Consider the identity map $I : (V, \|.\|) \to (V, \|.\|)$, where V is a normed linear space V. If for some $\alpha, \beta > 0$, $\beta\|x\| \leq \|x\|' \leq \alpha\|x\|$, for all $x \in V$, it implies that both the maps I and I^{-1} are continuous. It follows that the identity map I is a homeomorphism between $(V, \|.\|)$ and $(V, \|.\|')$. In the language of functional analysis, we call a bijective linear map $T : V \to V$ an *isomorphism* if both T and T^{-1} are continuous.

6.1.1 Finite dimensionality implies all norms equivalent

Theorem 6.1.1.1. *All norms are equivalent in a finite dimensional normed linear space.*

Proof. Let V be a finite dimensional normed linear space. To prove this theorem, it is enough to show that for any two norms $\|.\|$ and $\|.\|'$ on V, the identity map $I : (V, \|.\|) \to (V, \|.\|')$ is an isomorphism. Let us suppose dim $V = N$. Let, $\{v_1, \ldots, v_N\}$ be the basis of V. Let, $\{e_1, \ldots, e_N\}$ be the standard basis o $\mathbb{R}^N$. Consider the linear map

$$\begin{aligned} T : (\mathbb{R}^N, \|.\|_1) &\to (V, \|.\|) \\ e_i &\mapsto v_i. \end{aligned}$$

Then, for any $x \in \mathbb{R}^N$, $T(x) = \sum_{i=1}^{N} x_i v_i$. Since, T is a linear map which takes basis to basis, T is bijective. If we can show that T is an isomorphism, we will be done, because the composition of the continuous maps $T \circ T^{-1} = Id_V$ is an isomorphism.

Now, $\|T(x)\| = \|\sum_{i=1}^{N} x_i v_i\| \leq \max\limits_{1 \leq i \leq n} \|v_i\|\|x\|_1$, so the continuity of T when V is equipped with the $\|.\|$ norm is clear. Also, in the definition of the linear map T above, if we consider the codomain of T as V equipped with $\|.\|'$ norm, then with similar calculations, the continuity of T follows in this case too. Now, we will show the continuity of $T^{-1} : (V, \|.\|)(\text{or}, (V, \|.\|')) \to (\mathbb{R}^N, \|.\|_1)$. It is enough to check the continuity at $x = 0 \in V$. We proceed by contradiction. Let us assume T^{-1} is not continuous wrt $\|.\|$ norm. Then, there exists a sequence $\{y_n\}$ in $(V, \|.\|)$ such that $y_n \to 0$, but, for some $\epsilon > 0$, $\|T^{-1}(y_n)\| \geq \epsilon > 0$, for all $n \in \mathbb{N}$. Using, this sequence $\{y_n\}$, we define a new sequence $z_n = \frac{y_n}{\|T^{-1}(y_n)\|}$. Note that $z_n \to 0$ and $\|T^{-1}(z_n)\| = 1$, for all $n \in \mathbb{N}$. Let, $B = \{x \in \mathbb{R}^N : \|x\|_1 = 1\}$ denote the unit ball in $\mathbb{R}^N$, and we know that B is sequentially compact in $\mathbb{R}^N$. Since, for all $n \in \mathbb{N}$, $T^{-1}(z_n) \in B$, it admits a convergent subsequence $T^{-1}(z_{n_k})$ converging to let say, $y \in B$. Since, T is continuous, $T(T^{-1}(z_{n_k})) \to T(y)$, which implies $T(y) = 0$, since, $z_{n_k} \to 0$. But, T being injective implies that $y = 0$, contradicting the fact that $\|y\|_1 = 1$. Thus, T^{-1} is continuous. □

Remark 6.1.1.2. This type of proof is called the compactness-uniqueness method.

Corollary 6.1.1.3. *Any finite dimensional normed linear space is complete.*

Proof. Considering a finite dimensional vector space $(V, \|.\|)$, we have the above defined isomorphism $T : (\mathbb{R}^N, \|.\|_1) \to (V, \|.\|)$. Let, $\{y_n\}$ be a Cauchy sequence in V, then, $\{T^{-1}(y_n)\}$ is Cauchy sequence in $\mathbb{R}^N$. Since, $\mathbb{R}^N$ is complete, $\{T^{-1}(y_n)\}$ converges to say, $y \in \mathbb{R}^N$, and thus, $y_n \to T(y) \in V$. □

Corollary 6.1.1.4. *Let $(X, \|.\|)$ be a normed linear space. Let Y be a finite dimensional subspace of X. Then, Y is a closed subspace of X.*

Proof. From the previous corollary, since, Y is complete, it follows that Y is a closed subspace of X. □

6.1.2 Finite dimensionality implies compactness of unit ball

Theorem 6.1.2.1. *Let $(V, \|.\|)$ be a finite dimensional space. Then, the closed unit ball B is compact in V.*

Proof. Again, we consider the linear isomorphism $T : (\mathbb{R}^n, \|.\|_1) \to (V, \|.\|)$. Since, B is closed in V, $T^{-1}(B)$ being an inverse image of a closed set under a continuous map, is closed in $\mathbb{R}^N$. Also, $T^{-1}(B)$ is bounded by the operator norm on $\|T\|$ which is finite, since T is continuous. Therefore, $T^{-1}(B)$ is compact in $\mathbb{R}^N$, and thus, B being the continuous image of a compact set, is compact in V. □

Let us see some examples in infinite dimensional spaces.

Example 6.1.2.2. It is easy to see that the following two norms on $L^2(0, 1)$ are equivalent:

$$\|f\|_{L^2(0,1)} = \left(\int_0^1 |f|dx\right)^{1/2},$$

and,

$$\|f\|'_{L^2(0,1)} = \left(\int_0^1 w(x)|f|dx\right)^{1/2},$$

where $0 < m < w(x) < M$, for all $x \in (0, 1)$.

Clearly, for all $x \in (0, 1)$, $m\|f\|_{L^2(0,1)} \leq \|f\|' \leq M\|f\|_{L^2(0,1)}$.

Example 6.1.2.3. Consider the sup norm $\|.\|_\infty$ and the L^1 norm $\|.\|_1$ on $C[0, 1]$. For any $f \in C[0, 1]$, we always have $\|f\|_1 \leq \|f\|_\infty$. But, the other inequality doesn't hold, because if it were to hold, then there should exist some $\alpha > 0$ such that $\alpha\|f\|_\infty \leq \|f\|_1$, for all $f \in C[0, 1]$. But, considering our same old sequence of functions,

$$\begin{aligned} f_n &: [0, 1] \to \mathbb{R} \\ x &\mapsto \begin{cases} 1 - nx & \text{if } 0 \leq x \leq \frac{1}{n} \\ 0 & \text{if } \frac{1}{n} \leq x \leq 1. \end{cases} \end{aligned}$$

we should have $\alpha \leq \frac{1}{2n}$, for all $n \in \mathbb{N}$, and obviously, such an α cannot exist. So, the sup norm is stronger than the L^1 norm.

6.1.3 Equivalence of all norms implies finite dimensionality

Before we move on to proving what is mentioned in the title, let us see a lemma which will help us in proving it.

Lemma 6.1.3.1. *If X is an infinite dimensional normed linear space, and Y is any non-empty normed linear space, then, there exists a discontinuous linear map from X to Y.*

Proof. Since, X is an infinite dimensional vector space, we can choose a linearly independent set $A = \{a_1, a_2, \ldots, a_n, \ldots\}$. Let, $L = \{\frac{a_1}{\|a_1\|}, \frac{a_2}{2\|a_2\|}, \ldots, \frac{a_n}{n\|a_n\|}, \ldots\}$. Then, clearly L is a linearly independent set, and $0 \notin L$. So, we can extend L to a Hamel basis, say, B. Now, since, $Y \neq \{0\}$, we can choose $c(\neq 0) \in Y$.

Define

$$\begin{aligned} F &: X \longrightarrow Y \\ x &\longmapsto \begin{cases} c & \text{if } x \in L \\ 0 & \text{if } x \in B \setminus L. \end{cases} \end{aligned}$$

and then, we extend F linearly to the other parts of X. If we let $x_n = \frac{a_n}{n\|a_n\|}$, then, $\|x_n\| = \frac{1}{n} \to 0$ as $n \to \infty$, but, $F(x_n) = c \neq 0$ for all $n \in \mathbb{N}$. Thus, $F(x_n) \nrightarrow F(0)$, which implies F is not continuous. □

Theorem 6.1.3.2. *If V is a normed linear space in which all norms are equivalent, then, V is finite dimensional.*

Proof. If we go by the contrapositive method, then, we need to show that if V is infinite dimensional, then, all norms on V are not equivalent, in particular, it is enough to show the existence of any two non-equivalent norms on V.

Let, $\|.\|$ be a norm defined on V. From the above lemma, there exists a discontinuous linear functional $f : V \to \mathbb{R}$. Let us define a new norm $\|.\|_*$ on V, where

$$\|x\|_* = \|x\| + |f(x)|,$$

for all $x \in V$. It is easy to prove that $\|.\|$ indeed defines a norm on V. Now, if $\|.\|$ and $\|.\|_*$ are equivalent, then, there should exist positive reals α_1 and α_2 such that $\alpha_1\|x\| \leq \|x\|_* \leq \alpha_2\|x\|$. By definition of the norm $\|.\|_*$, we have $\|x\| \leq \|x\|_*$, so we can take $\alpha_1 = 1$. Now, let us suppose there exists α_2 such that $\|x\|_* \leq \alpha_2\|x\|$. Then, we have that $|f(x)| \leq (\alpha_2 - 1)\|x\|$, but this implies that f is a continuous linear map, a contradiction. So, the norms $\|.\|$ and $\|.\|_*$ are not equivalent on V. ☐

Lemma 6.1.3.3. ***(Riesz Lemma).*** *Let V be a normed linear space and W be a closed proper subspace of V ($V \neq W$). Then, for every $0 < \epsilon < 1$, there exists a $u_\epsilon \in V$ such that*

i. $\|u_\epsilon\| = 1$,

ii. $1 - \epsilon \leq d(u_\epsilon, W) \leq 1$,

where $d(u_\epsilon, W) = \underset{y\in W}{inf}\|u_\epsilon - y\|$.

Proof. Since, W is a closed subspace, we can always find a $v \in V$ such that $d(v, W) = \delta$, for some $\delta > 0$. Now, for every $0 < \epsilon < 1$, we have $\delta < \frac{\delta}{1-\epsilon}$, hence, there exists a $w_\epsilon \in W$ such that $\delta \leq \|v - w_\epsilon\| \leq \frac{\delta}{1-\epsilon}$. We define

$$u_\epsilon = \frac{v - w_\epsilon}{\|v - w_\epsilon\|}.$$

Consider any $z \in W$. Then,

$$\begin{aligned}\|u_\epsilon - z\| &= \left\|\frac{v - w_\epsilon}{\|v - w_\epsilon\|} - z\right\| \\ &= \frac{\|v - (w_\epsilon + z\|v - w_\epsilon\|)\|}{\|v - w_\epsilon\|} \\ &\geq \frac{1-\epsilon}{\delta} \times \delta = 1 - \epsilon.\end{aligned}$$

Also, since, $0 \in W$, $\|u_\epsilon - 0\| = \|u_\epsilon\| = 1$. Hence, for all $z \in W$, $1 - \epsilon \leq \|u_\epsilon - z\| \leq 1$, that is, $1 - \epsilon \leq d(u_\epsilon, W) \leq 1$. ☐

Remark 6.1.3.4. If V is a finite dimensional normed linear space, then, there exists $u \in V$ such that $\|u\| = 1$ and $d(u, W) = 1$.

From Riesz lemma, we have that for every $0 < \epsilon < 1$, $0 < 1-\epsilon \leq \inf_{y\in W}\|u_\epsilon - y\| \leq 1$. Since, V is finite dimensional, the unit ball in V is compact. Hence, letting $\epsilon \to 0$, since $\|.\|$ is a continuous function, there exits $y \in W$ such that $d(y, W) = 1$.

Note: For Hilbert spaces, we can show that there exists $u \in V$ such that $\|u\| = 1$ and $d(u, W) = 1$.

6.1.4 Compactness of unit ball implies finite dimensionality

Theorem 6.1.4.1. *Let the unit ball $B = \{x \in V : \|x\| = 1\}$ be compact in the normed linear space V. Then, V is finite dimensional.*

Proof. The compactness of B implies that B is totally bounded. Hence, for every $\delta > 0$, there exists $x_1, x_2, \ldots, x_n \in B$ such that $B \subseteq \cup_{i=1}^n B(x_i, \delta)$. Let us define

$$W = span\{x_1, x_2, \ldots, x_n\}, \tag{5}$$

which implies dim $W \leq n$. If we can show $V = W$, we will be done, as then V will be a finite dimensional space.

Let us suppose W is a proper subspace of V. Now, since, W is a finite dimensional subspace of V, it implies W is a closed subspace of V. So, using Riesz lemma, while considering $\epsilon = \frac{1}{3}$, we have that there exists $u_{\frac{1}{3}} \in V$ such that $\|u_{\frac{1}{3}}\| = 1$, and $d(u_{\frac{1}{3}}, W) \geq \frac{2}{3}$. Since, $\|u_{\frac{1}{3}}\| = 1$, it implies $u_{\frac{1}{3}} \in B$, and hence, from equation 5, considering $\delta = \frac{1}{2}$, it follows that $\|u_{\frac{1}{3}} - x_i\| < \frac{1}{2}$, for some i. But, we also have that $\|u_{\frac{1}{3}} - x_i\| \geq \frac{2}{3}$, for all i, which gives a contradiction. $\square$

7 Quotient spaces

As one might have noticed, we have emphasized a lot on closed subspaces of normed linear spaces, and this is the section where one can finally start appreciating the importance of existence of such spaces. So, let's start!

Let X be a vector space and Y be a proper subspace of X. We consider our usual quotient vector space X/Y.

Define $\sim$ on X, where $x \sim y$ if $x - y \in Y$, and consequently,

$$X/Y := \{x + Y : x \in X\},$$

on which we consider our usual addition and scalar multiplication.

Now, assuming $(X, \|.\|)$ to be a normed linear space and Y to be a closed proper subspace of X, we would like to define a norm on the quotient space X/Y. We will soon see why the closedness of Y is required.

For any $x + Y \in X/Y$, we define

$$|||x + Y|||_{X/Y} = \inf\{\|x + y\| : y \in Y\}.$$

It is clear from the definition, that the norm gives the distance of $x \in X$ from the set Y. Also, one should check,though it's an easy observation, the well definedness of this definition by showing that if $x \sim x'$, then, $|||x + Y||| = |||x' + Y|||$.

Now, using this definition, we show that the space $(X/Y, |||.|||_{X/Y})$ is indeed a normed linear space.

i. By definition, positivity of the norm $|||.|||_{X/Y}$ is clear. We would like to show that if $|||x+Y|||_{X/Y} = 0$, then, $x \in Y$. We can consider a sequence y_n such that $\|x + y_n\| \to 0$, which implies $y_n \to -x$. Since, Y is closed, it follows that $-x \in Y$, that is, $x \in Y$, because Y is a subspace of X.

   NOTE : The above would not have been true if Y was not a closed subspace of X, henceforth, the existence of such a norm on the quotient space X/Y it totally dependent on the closedness of Y as a subspace of X.

ii. Now, we check the homogeneity property of $|||.|||_{X/Y}$. Let, us consider $k \neq 0$. Then,

$$\begin{aligned}
|||kx + Y|||_{X/Y} &= \inf\{\|kx + y\| : y \in Y\} \\
&= \inf\{|k|\|x + y/k\| : y \in Y\} \\
&= \inf\{\|x + z\| : z \in Y\} \\
&= \inf|k|\{\|x + z\| : z \in Y\} \\
&= |k||||x + Y|||_{X/Y}.
\end{aligned}$$

iii. We are just left with showing the triangle inequality. We will show that $|||(x + z) + Y|||_{X/Y} \leq |||x + Y|||_{X/Y} + |||z + Y|||_{X/Y}$, where $x, z \in X$. By definition of infimum, there exists y_1, y_2 such that $\|x + y_1\| \leq |||x + Y|||_{X/Y} + \epsilon/2$, and $\|z + Y\| \leq |||z + Y|||_{X/Y} + \epsilon/2$. Therefore,

$$|||(x + z) + Y|||_{X/Y} \leq \|x + z + (y_1 + y_2)\| \leq |||x + Y|||_{X/Y} + |||z + Y|||_{X/Y}.$$

Lemma 7.0.0.1. *If $\{x_n + Y\}$ is a sequence in X/Y such that $x_n + Y \to x + Y$, the, there exists some sequence $\{y_n\} \in Y$ such that $x_n + y_n \to x$ in X. Conversely, if for some $\{y_n\}$ in Y, $x_n + y_n \to x$ in X, then, $x_n + Y \to x + Y$ in X/Y.*

Proof. The proof uses the same ideas we used to proof that $|||.|||_{X/Y}$ is a norm. So, we omit the proof. $\square$

Theorem 7.0.0.2. *Let $(X, \|.\|)$ be a normed linear space and Y be a closed subspace of X. Then, X is a Banach space iff Y and X/Y are Banach spaces.*

Proof. We first prove the forward direction. Since, X is a Banach space, and Y is a closed subspace of X, using the fact that closed subspaces of complete metric spaces are complete, we get that Y is complete, that is, Y is a Banach space. Now, to show that X/Y is a Banach space, we consider a Cauchy sequence $\{x_n + Y\}$ in X/Y. Using the Cauchyness of this sequence, we choose one of its subsequence $\{x_{n_k} + Y\}$ which satisfies

$$|||(x_{n_{k+1}} + Y) - (x_{n_k} + Y)|||_{X/Y} < \frac{1}{2^k}$$

for all $k \in \mathbb{N}$. Choosing $y_1 = 0$, we have

$$\inf_{y \in Y} |||(x_{n_2} - y) - (x_{n_1} - y_1)|||_{X/Y} < \frac{1}{2}.$$

Since, Y is closed, there exists $y_2 \in Y$ such that

$$|||(x_{n_2} - y_2) - (x_{n_1} - y_1)|||_{X/Y} < \frac{1}{2}.$$

Now, as $y_2 \in Y$, we have

$$\inf_{y \in Y} |||(x_{n_3} - y) - (x_{n_2} - y_2)|||_{X/Y} < \frac{1}{2^2},$$

which again implies there exists $y_3 \in Y$ such that

$$|||(x_{n_3} - y_3) - (x_{n_2} - y_2)|||_{X/Y} < \frac{1}{2^2}.$$

Continuing, there exists $y_{k+1} \in Y$ such that

$$|||(x_{n_{k+1}} - y_{k+1}) - (x_{n_k} - y_k)|||_{X/Y} < \frac{1}{2^k}.$$

Now, let us define $z_k = x_{n_k} - y_k$. From the above, with some easy proof, it follows that z_k is a Cauchy sequence in X, and since X is a Banach space, $z_k \to z \in X$, that is, $x_{n_k} + (-y_k) \to z$ as $k \to \infty$. From lemma 7.0.0.1, we have that $x_{n_k} + Y \to z + Y$ as $k \to \infty$, and thus, $x_n + Y \to z + Y$. Therefore, X/Y is a Banach space.

Now, we proof the backward direction, that is, considering Y and X/Y are Banach spaces, we will show that X is a Banach space. Consider a Cauchy sequence $\{x_n\}$ in X. So, for $\epsilon > 0$ be arbitrary, there exists $N(\epsilon) \in \mathbb{N}$, such that

$$\|x_n - x_m\| < \epsilon/2, \quad \text{for all } m, n \geq N(\epsilon).$$

Therefore, from the definition of infimum, since

$$|||(x_n + Y) - (x_m + Y)|||_{X/Y} \leq \|x_n - x_m\|,$$

it follows that $\{x_n + Y\}$ is a Cauchy sequence in X/Y. Since, X/Y is a Banach space, $x_n + Y \to x + Y$, for some $x + Y \in X/Y$, and then, from 7.0.0.1, there exists a sequence $\{y_n\} \in Y$ such that $x_n + y_n \to x \in X$. Now, we claim that $\{y_n\}$ is a Cauchy sequence in Y. Indeed,

$$\|y_n - y_m\| \leq \|(x_n + y_n) - (x_m + y_m)\| + \|x_m - x_n\| < \epsilon,$$

as $\{x_n + y_n\}$ is convergent, hence a Cauchy sequence. Since, Y is a Banach space, and we got that $\{y_n\}$ is Cauchy, there exists $y \in Y$ such that $y_n \to y$. So, we have that $x_n \to x - y \in X$, and thus, X is a Banach space. $\square$